

Data Mining – Lecture 8 & 9 – Classification Part 2 & 3

1-What is the primary characteristic of a Naïve Bayesian classifier?

- a- It predicts the exact class of a tuple.
- b- It predicts the probability of a tuple belonging to a specific class.
- c- It relies solely on attributes' effect on class determination.
- d- It uses a complex mathematical formula for classification.

Answer : b.

2- What is the basis of the Naïve Bayesian classifier?

- a- The Central Limit Theorem
- b- The Pythagorean Theorem
- c- Bayes Theorem
- d- Newton's Laws of Motion

Answer : c.

Note “ naïve refers to the classifier which assumes independence among attributes .

3- We use Laplacian correction to fix the Zero problem , which we replace the zero values with a non-zero value but close to zero .

- a- True
- b- False

Answer : a.

$C1(\text{Senior}) = 80$, $C2(\text{Junior}) = 120$ $P(C1)=80/200$ $P(C2)=120/200$
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X: {department = “marketing”; age= “youth”, salary= low}

4- What does the term "C1 (Senior) = 80" represent in this context?

- a- The probability of tuple X belonging to the "Senior" class.
- b- The count of instances classified as "Senior".
- c- The probability of encountering a "Senior" tuple in the dataset.
- d- The number of attributes associated with the "Senior" class.

Answer : b.

5- What is the value of $P(X|C2)P(C2)$ for the given Tuple X?

- a- 0.055
- b- 0.033
- c- 0.022
- d- 0.000

Answer : b.

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6- Based on the provided calculations, what is the probability of tuple X belonging to class "Senior"?

- a- 0
- b- 0.055
- c- 0.033
- d- 0.022

Answer : a , note “ notice that this question and the previous one are the same but in different forms”

7- What is the value of $P(X|C2)P(C2)$ for the given Tuple X after applying the Laplacian estimator?

- a- 0.055
- b- 0.033
- c- 0.025
- d- 0.00002

Answer : b , note “ it’s the same answer as before , why ? because all the values are non-zero “

8- Based on the provided calculations with the Laplace estimator, what is the Laplace-estimated probability of tuple X belonging to class “Senior”?

- a- 0
- b- 0.055
- c- 0.033
- d- 0.000008

Answer : d , note “ take a look back at question 6 “

9- What is a characteristic feature of Lazy Learners ?

- a- They immediately classify test data upon receiving it.
- b- They delay classification until new test data is available.
- c- They only consider a single attribute for classification.
- d- They store all training data in memory permanently.

Answer : b.

10- What role does k play in K-Nearest Neighbor (KNN) classifiers?

- a- It denotes the number of attributes used for classification.
- b- It represents the distance metric used for computing similarity.
- c- It specifies the range of values for normalization.
- d- It indicates the number of closest neighbors considered for classification.

Answer : d.

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11- Why is normalization necessary in KNN classifiers?

- a- To reduce the computational complexity of distance calculations.
- b- To eliminate outliers from the training data.
- c- To ensure that all attributes contribute equally to the distance measure.
- d- To adjust for variations in the ranges of different attributes.

Answer : d.

12- How is the class of a test data tuple determined in KNN classifiers?

- a- By selecting the class label of the nearest training data tuple.
- b- By performing logistic regression on the training data.
- c- By calculating the mean value of all class labels in the training data.
- d- By majority voting of the class labels of the k closest neighbors.

Answer : d.

13- What is the primary purpose of creating a scatter plot before performing linear regression analysis?

- a- To visualize the relationship between the dependent and independent variables.
- b- To calculate the slope and intercept of the regression line.
- c- To determine the statistical significance of the regression model.
- d- To identify outliers in the dataset.

Answer : a.

14- In the linear regression equation $Y = a + bX$ what does the variable Y represent ?

- a- The slope of the line.
- b- The dependent variable.
- c- The y-intercept of the line.
- d- The independent variable.

Answer : b.

15- b in linear regression equation $Y = a + bX$ represents the slope of the line .

- a- True
- b- False

Answer : a.

-----Lec 9 -----

16- What are "Positives" in the context of evaluating classifier performance?

- a- Tuples representing the class of interest.
- b- Tuples representing other class(es).
- c- Tuples correctly labeled by the classifier.
- d- Tuples incorrectly labeled by the classifier.

Answer : a.

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17- What do "True Positives" represent in model evaluation metrics?

- a- Tuples representing the class of interest.
- b- Tuples representing other class(es).
- c- Positive tuples incorrectly labeled by the classifier.
- d- Positive tuples correctly labeled by the classifier.

Answer : d.

18- How are "False Positives" defined in classifier performance evaluation?

- a- Tuples representing the class of interest.
- b- Tuples representing other class(es).
- c- Negative tuples incorrectly labeled by the classifier.
- d- Positive tuples correctly labeled by the classifier.

Answer : c.

19- What do "True Negatives" represent in model evaluation metrics?

- a- Tuples representing the class of interest.
- b- Tuples representing other class(es).
- c- Negative tuples incorrectly labeled by the classifier.
- d- Negative tuples correctly labeled by the classifier.

Answer : d.

20- How are "False Negatives" defined in model evaluation metrics?

- a- Tuples representing the class of interest.
- b- Tuples representing other class(es).
- c- Negative tuples incorrectly labeled by the classifier.
- d- Positive tuples incorrectly labeled by the classifier.

Answer : d .

21 - Which type of cross-validation is recommended class distribution in each fold is approximately the same as in initial dataset?

- a- Holdout
- b- Random Subsampling
- c- Stratified k-fold cross-validation
- d- k-fold cross-validation

answer : c. note “ stratified 10 fold cross is recommended “

22- What does an ensemble method consist of in the context of improving classification accuracy?

- a- A set of classifiers, each with a vote for a class label.
- b- A single classifier trained on the entire dataset.
- c- Multiple classifiers with the same parameters.
- d- A classifier trained on a single partition of the dataset.

Answer : a.

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23- What method is commonly used to compose an aggregate classification in ensemble methods?

- a- Mean voting
- b- Median voting
- c- Majority voting
- d- Weighted voting

Answer : c.

"شكرا لكم ولا تنسونا بالدعاء"